Empirical Proof Record: VALIDATED

2026-05-22T17:09:46.396192+00:00

Empirical Proof Record — VALIDATED

Proof ID: `a63db28704b781f2ffd08558719d4446…`
Created: 2026-05-22T17:09:46.396192+00:00

1. Claim

When an identically-recognised probe is re-exposed across a trajectory, it lands on a strictly-deeper (more-scarred) manifold every time: memory_path_dependence = (recognised re-exposure transitions with Δ permanent-scars > 0) / (recognised re-exposure transitions) >= 1.0, with the whole-run scar count monotonic non-decreasing (a scar cannot be reset). This measures memory where Kimera keeps it — the scars and the S4 manifold — not the content-recognition layer.

- **Operationalisation:** Run 8 stimuli each re-exposed 3x, interleaved (gap = len(stimuli)), through Kimera's entity target (Takwin) in batch mode so the substrate accumulates across the batch. Per cycle read vault_stats.total_scars_stored (canonical permanent scars), arachne_web_coupling_frobenius (manifold deformation), and the concept set (recognition). For each same-stimulus consecutive exposure pair: recognised iff Jaccard(concepts) >= 0.80; confirmed iff recognised AND Δ scars > 0. memory_path_dependence = confirmed / recognised. Forced to 0.0 (REFUTED) if the whole-run scar count ever decreases.
- **Threshold:** `memory_path_dependence >= 1.0 proportion`
- **H0:** H0: memory_path_dependence < 1.0 — some re-exposure of a recognised probe did NOT land on more permanent scars (the substrate re-derives without accumulating), or a scar was reset; not path-dependent permanent memory
- **H1:** H1: memory_path_dependence >= 1.0 — every re-exposure of a recognised probe lands on a strictly-deeper manifold and no scar is ever reset; path-dependent permanent memory in the scar/S4 substrate

2. Verdict

VALIDATED — observed `1.0`
memory_path_dependence 1.0000 = 16/16 recognised re-exposure transitions landed on strictly more permanent scars; 0 transitions failed recognition (excluded). Permanence: HOLDS (0 scar-count decreases over the run). Scars 1.0→24.0; coupling 1.293702→10.47013; phi |delta| mean 0.0000 (Φ-rigid → memory is in the scars/manifold, not Φ); scar-depth vs exposure-ordinal Spearman rho=0.9436, p=2.4e-12 (deepens with experience)

3. Evidence

Pillar	Statistic	Value	95% CI	Library
`O.flow.memory_path_dependence`	`memory_path_dependence`	`1.0`	—	ophamin 0.109.0

4. Pre-registration

- Registered at: `2026-05-22T17:09:46.139642+00:00`
- Config hash: `f5c0c3ae55bb8ed69287755ab991315a…`
- Data hash: `4e6090352a40ea27df1cf08fb4d02bd5…`

5. Signature

`7bc3d8aed9ec1648ba5bb56d744be7458106bb98812314fd33fc56eafe0a7b5d`